

1. Entities, Relationships, and Attributes

- PERSON: Ssn, Fname, Minit, Lname, Bdate, Sex, No, Street, Apt_no, City, State, Zip
- FACULTY: Rank, Fphone, Office, Salary
- STUDENT: Class
- GRAD_STUDENT: Degrees (College, Degree, Year)
- GRANT: No, Title, Agency, St_date
- INSTRUCTOR_RESEARCHER: no attributes shown directly
- DEPARTMENT: Dname, Dphone, Office
- COLLEGE: Cname, Dean, Coffice
- COURSE: C#, Cname, Cdesc
- SECTION: Sec#, Year, Qtr
- CURRENT_SECTION: derived from SECTION where quarter/year are current

Relationships seen in the diagram:

- ADVISOR between FACULTY and GRAD_STUDENT (1:N)
- COMMITTEE between FACULTY and GRAD_STUDENT (M:N)
- PI between FACULTY and GRANT (1:N)
- SUPPORT between GRANT and INSTRUCTOR_RESEARCHER (M:N), with Start, Time, End
- BELONGS between FACULTY and DEPARTMENT (M:1)
- CHAIRS between FACULTY and DEPARTMENT (1:1)
- TEACH between INSTRUCTOR_RESEARCHER and DEPARTMENT (1:N as drawn)
- MAJOR between STUDENT and DEPARTMENT (N:1)
- MINOR between STUDENT and DEPARTMENT (N:1)
- REGISTERED between STUDENT and CURRENT_SECTION (M:N)

- TRANSCRIPT between STUDENT and SECTION (M:N), with Grade
- CD between COLLEGE and DEPARTMENT (1:N)
- DC between DEPARTMENT and COURSE (1:N)
- CS between COURSE and SECTION (1:N)

2. How I handled inheritance

I used one table for PERSON and separate tables for FACULTY, STUDENT, and GRAD_STUDENT. The subtype tables use the same primary key as PERSON, so their PK is also an FK back to PERSON. This works well for specialization and keeps shared person data in one place.

For PERSON -> FACULTY/STUDENT, I treated it as disjoint because the diagram shows a d. For STUDENT -> GRAD_STUDENT, I made GRAD_STUDENT a subtype of STUDENT.

For INSTRUCTOR_RESEARCHER, I treated it as a category/union of FACULTY and GRAD_STUDENT. I gave it its own key and allowed it to point to either a faculty member or a grad student.

3. Tables

PERSON

- ssn CHAR(9) PRIMARY KEY
- fname VARCHAR(30)
- minit CHAR(1)
- lname VARCHAR(30)
- bdate DATE
- sex CHAR(1)

- addr_no VARCHAR(10)
- street VARCHAR(50)
- apt_no VARCHAR(10)
- city VARCHAR(30)
- state CHAR(2)
- zip VARCHAR(10)

FACULTY

- ssn CHAR(9) PRIMARY KEY
- rank VARCHAR(30)
- fphone VARCHAR(15)
- office VARCHAR(20)
- salary DECIMAL(10,2)
- dept_name VARCHAR(50) NOT NULL
- FK ssn -> PERSON(ssn)
- FK dept_name -> DEPARTMENT(dname)

STUDENT

- ssn CHAR(9) PRIMARY KEY
- class INT
- major_dept_name VARCHAR(50) NOT NULL
- minor_dept_name VARCHAR(50)
- FK ssn -> PERSON(ssn)
- FK major_dept_name -> DEPARTMENT(dname)
- FK minor_dept_name -> DEPARTMENT(dname)

GRAD_STUDENT

- ssn CHAR(9) PRIMARY KEY
- advisor_ssn CHAR(9)
- FK ssn -> STUDENT(ssn)
- FK advisor_ssn -> FACULTY(ssn)

GRAD_STUDENT_DEGREE

- grad_student_ssn CHAR(9)
- college VARCHAR(50)
- degree VARCHAR(50)
- year INT
- PRIMARY KEY (grad_student_ssn, college, degree, year)
- FK grad_student_ssn -> GRAD_STUDENT(ssn)

GRANT_PROJECT

- grant_no INT PRIMARY KEY
- title VARCHAR(100)
- agency VARCHAR(100)
- st_date DATE
- pi_ssn CHAR(9) NOT NULL
- FK pi_ssn -> FACULTY(ssn)

INSTRUCTOR_RESEARCHER

- ir_id INT PRIMARY KEY
- faculty_ssn CHAR(9)

- grad_student_ssn CHAR(9)
- dept_name VARCHAR(50) NOT NULL
- FK faculty_ssn -> FACULTY(ssn)
- FK grad_student_ssn -> GRAD_STUDENT(ssn)
- FK dept_name -> DEPARTMENT(dname)

SUPPORT

- grant_no INT
- ir_id INT
- start_date DATE
- support_time DECIMAL(5,2)
- end_date DATE
- PRIMARY KEY (grant_no, ir_id)
- FK grant_no -> GRANT_PROJECT(grant_no)
- FK ir_id -> INSTRUCTOR_RESEARCHER(ir_id)

COMMITTEE

- faculty_ssn CHAR(9)
- grad_student_ssn CHAR(9)
- PRIMARY KEY (faculty_ssn, grad_student_ssn)
- FK faculty_ssn -> FACULTY(ssn)
- FK grad_student_ssn -> GRAD_STUDENT(ssn)

COLLEGE

- cname VARCHAR(50) PRIMARY KEY
- dean VARCHAR(60)

- coffice VARCHAR(20)

DEPARTMENT

- dname VARCHAR(50) PRIMARY KEY
- dphone VARCHAR(15)
- office VARCHAR(20)
- college_name VARCHAR(50) NOT NULL
- chair_ssn CHAR(9) UNIQUE
- FK college_name -> COLLEGE(cname)
- FK chair_ssn -> FACULTY(ssn)

COURSE

- course_no VARCHAR(15) PRIMARY KEY
- cname VARCHAR(60)
- cdesc VARCHAR(255)
- dept_name VARCHAR(50) NOT NULL
- FK dept_name -> DEPARTMENT(dname)

SECTION

- sec_no INT
- year INT
- qtr VARCHAR(10)
- course_no VARCHAR(15) NOT NULL

- PRIMARY KEY (sec_no, year, qtr)
- FK course_no -> COURSE(course_no)

CURRENT_SECTION

- sec_no INT
- year INT
- qtr VARCHAR(10)
- PRIMARY KEY (sec_no, year, qtr)
- FK (sec_no, year, qtr) -> SECTION(sec_no, year, qtr)

REGISTERED

- student_ssn CHAR(9)
- sec_no INT
- year INT
- qtr VARCHAR(10)
- PRIMARY KEY (student_ssn, sec_no, year, qtr)
- FK student_ssn -> STUDENT(ssn)
- FK (sec_no, year, qtr) -> CURRENT_SECTION(sec_no, year, qtr)

TRANSCRIPT

- student_ssn CHAR(9)
- sec_no INT
- year INT
- qtr VARCHAR(10)
- grade VARCHAR(2)

- PRIMARY KEY (student_ssn, sec_no, year, qtr)
- FK student_ssn -> STUDENT(ssn)
- FK (sec_no, year, qtr) -> SECTION(sec_no, year, qtr)

4. Which relationships became foreign keys and which became separate tables

Relationships turned into foreign keys: BELONGS, ADVISOR, PI, CD, DC, CS, MAJOR, MINOR, CHAIRS, and TEACH. These are mostly 1:N or 1:1 relationships, so adding a foreign key was enough.

Relationships turned into separate tables: COMMITTEE, SUPPORT, REGISTERED, and TRANSCRIPT. These are M:N relationships, and some also have their own attributes like Grade or Start/End/Time.

5. Multi-valued attribute handling

The attribute Degrees for GRAD_STUDENT is multi-valued and composite, so I could not store it in one column. I created a separate table called GRAD_STUDENT_DEGREE with columns for grad_student_ssn, college, degree, and year. This lets one grad student have multiple previous degrees.

6. Assumptions

- PERSON specialization is disjoint because the diagram shows d.
- A separate table is used for each subtype instead of putting everything in one big table.
- INSTRUCTOR_RESEARCHER is a union/category from FACULTY and GRAD_STUDENT.
- CURRENT_SECTION could really be a view, but I listed it as a table since the assignment asks for tables.
- SECTION uses a composite key (sec_no, year, qtr) because section numbers may repeat in different terms.
- A student has one major and may have one minor.
- SUPPORT.Time was treated as a numeric amount of time.

7. Final list of tables

PERSON, FACULTY, STUDENT, GRAD_STUDENT, GRAD_STUDENT_DEGREE,
GRANT_PROJECT, INSTRUCTOR_RESEARCHER, SUPPORT, COMMITTEE, COLLEGE,
DEPARTMENT, COURSE, SECTION, CURRENT_SECTION, REGISTERED,
TRANSCRIPT